
INDUSTRIAL INTERNSHIP REPORT

Food Delivery Management System using Core Java

Submitted in Partial Fulfillment of the Requirements for the Industrial Internship

Submitted By

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B.Tech Computer Science & Engineering

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Shahpur, Jalandhar

Internship Organization

UniConverge Technologies Pvt. Ltd.

Through

Upskill Campus

Internship Duration

09 June 2026 – 12 July 2026

CERTIFICATE

This is to certify that Mr. Akshey Mahajan, a student of B.Tech Computer Science & Engineering (Second Year) at CT Institute of Technology, Shahpur, Jalandhar, has successfully completed his Industrial Internship titled “Food Delivery Management System using Core Java” under the guidance of UniConverge Technologies Pvt. Ltd. Through upskill Campus during the period from 09 June 2026 to 12 July 2026.

During the internship, the student actively participated in learning Core Java concepts, software development methodologies, object-oriented programming principles, and practical application development. The student completed the assigned project and demonstrated dedication, technical understanding, and enthusiasm throughout the internship period.

This report is submitted in partial fulfillment of the requirements for the Industrial Internship.

(Signature)

Department of Computer Science & Engineering

CT Institute of Technology

DECLARATION

I hereby declare that this internship report entitled “Food Delivery Management System using Core Java” is my original work carried out during the Industrial Internship conducted by UniConverge Technologies Pvt. Ltd. Through upskill Campus.

The work presented in this report has not been submitted previously for the award of any degree, diploma, or certification. All information obtained from books, websites, journals, and other sources has been acknowledged appropriately in the references section.

I understand that any form of plagiarism or misrepresentation may result in the rejection of this report.

Submitted By

Akshey Mahajan

B.Tech Computer Science & Engineering (Second Year)

CT Institute of Technology

Date: ____31/07/2007____

ACKNOWLEDGEMENT

I express my sincere gratitude to UniConverge Technologies Pvt. Ltd. And upskill Campus for providing me with the opportunity to undertake this Industrial Internship in Core Java.

This internship allowed me to gain practical exposure to software development and strengthened my understanding of object-oriented programming concepts. Working on the Food Delivery Management System helped me improve my programming skills, logical thinking, debugging techniques, and understanding of application development.

I also express my heartfelt thanks to the faculty members of CT Institute of Technology, Shahpur, Jalandhar, for their continuous encouragement, motivation, and support throughout the internship.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of this internship. Their guidance and encouragement played an important role in enhancing my technical knowledge and confidence.

EXECUTIVE SUMMARY

This report presents the work carried out during the six-week Core Java Industrial Internship organized by UniConverge Technologies Pvt. Ltd. In collaboration with upskill Campus.

The internship was designed to provide practical knowledge of software development using Core Java. During the internship, I learned various programming concepts including variables, data types, operators, control statements, methods, classes, objects, constructors, inheritance, polymorphism, encapsulation, abstraction, exception handling, collections, and basic file handling.

As a part of the internship, I developed a Food Delivery Management System using Core Java. The application simulates a simple online food ordering system where users can register, log in, browse restaurant menus, select food items, place orders, generate bills, and manage customer information. The project follows object-oriented programming principles and demonstrates the practical implementation of Core Java concepts.

The internship also enhanced my problem-solving skills, logical thinking, debugging ability, and understanding of software development methodologies. It provided valuable industry exposure and strengthened my confidence in designing Java-based applications.

Overall, this internship served as an important step toward my goal of becoming a Software Engineer. It motivated me to continue learning advanced Java technologies, Data Structures and Algorithms, and Full Stack Development to build scalable software applications in the future.

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Industrial internships are an essential part of engineering education as they bridge the gap between theoretical knowledge and practical implementation. They provide students with an opportunity to understand professional work culture, industry expectations, and real-world software development practices.

The Core Java Internship organized by UniConverge Technologies Pvt. Ltd. In collaboration with upskill Campus provided me with valuable exposure to programming concepts and software application development. Throughout the internship, I explored various Core Java concepts including object-oriented programming, exception handling, collections, file handling, and modular programming.

As part of the internship, I developed a Food Delivery Management System using Core Java. This project allowed me to apply theoretical concepts in a practical environment while strengthening my logical thinking and programming abilities. It also enhanced my confidence in developing console-based Java applications.

This report documents the learning experience, project development process, implementation methodology, testing procedures, and the knowledge gained during the internship.

INTRODUCTION

Technology has transformed almost every sector of society, and the food delivery industry is one of the fastest-growing digital service sectors. Customers now prefer ordering food online instead of visiting restaurants, making efficient software solutions increasingly important.

Software development plays a significant role in creating such applications. Core Java provides a strong foundation for building reliable, scalable, and object-oriented software systems. It introduces programmers to essential concepts such as classes, objects,

inheritance, polymorphism, encapsulation, collections, exception handling, and file management.

The objective of this internship was to strengthen programming skills by implementing these Core Java concepts through an industry-oriented project. The Food Delivery Management System was developed as a console-based application demonstrating the workflow of an online food ordering platform.

The project focuses on creating an organized application where users can browse restaurants, select food items, place orders, generate bills, and manage customer information using object-oriented programming principles.

ABOUT UNICONVERGE TECHNOLOGIES PVT. LTD.

UniConverge Technologies Pvt. Ltd. Is a technology-driven organization dedicated to providing innovative software solutions and industrial training programs. The company specializes in modern technologies including Java Development, Python, Artificial Intelligence, Machine Learning, Internet of Things (IoT), Cloud Computing, Cyber Security, Data Analytics, and Full Stack Development.

The organization focuses on bridging the gap between academic education and industrial requirements by offering internships, practical projects, technical workshops, and mentorship programs. These initiatives help students gain hands-on experience while improving their employability and technical competence.

Through structured learning modules and industry-oriented assignments, UniConverge Technologies enables students to understand professional software development methodologies and current industry practices.

ABOUT UPSKILL CAMPUS

Upskill Campus is a leading career development platform that collaborates with educational institutions and industry partners to provide students with practical learning opportunities.

The platform offers internships, certification programs, industrial projects, workshops, and mentoring sessions across multiple technical domains. Its primary objective is to enhance students' technical knowledge while preparing them for future careers in the software industry.

By participating in internships organized through upskill Campus, students gain practical exposure to software development, documentation, teamwork, communication skills, and project implementation methodologies.

OBJECTIVES OF THE INTERNSHIP

The major objectives of this internship are:

To strengthen Core Java programming skills.

To understand object-oriented programming concepts.

To develop logical and analytical thinking.

To gain practical experience in software application development.

To implement classes, objects, inheritance, and polymorphism.

To understand exception handling and file handling.

To improve debugging and testing skills.

To enhance documentation and reporting skills.

To gain exposure to industry-oriented software development practices.

To prepare for future opportunities in software engineering.

PROJECT INTRODUCTION

The Food Delivery Management System using Core Java is a console-based application designed to simulate the basic functionality of an online food ordering platform.

The primary objective of this project is to simplify food ordering by providing users with an organized system where they can browse restaurants, view available food items, place orders, calculate bills, and manage customer information.

The application has been developed using Core Java programming concepts while following object-oriented programming principles. Different modules are designed independently to improve code organization, maintainability, and scalability.

The project demonstrates the practical implementation of Java classes, objects, methods, constructors, collections, exception handling, conditional statements, loops, and file handling.

Although the current version is console-based, the application can be extended in the future by integrating graphical user interfaces, databases, payment gateways, and web technologies.

PROJECT FEATURES

The Food Delivery Management System provides the following features:

User Registration

User Login

Restaurant Menu Display

Food Item Selection

Order Placement

Quantity Selection

Bill Generation

Customer Information Management

Order Summary

Input Validation

Exception Handling

Modular Program Structure

PROJECT DESCRIPTION

The Food Delivery Management System using Core Java is a console-based software application designed to simplify the food ordering process by providing users with an organized and efficient ordering platform. The system enables customers to register, log in, browse restaurant menus, select food items, place orders, and generate bills through an interactive menu-driven interface.

The project has been developed using Core Java while following the principles of Object-Oriented Programming (OOP). Each module of the application has been designed independently to improve readability, maintainability, and future scalability.

The application demonstrates practical implementation of programming concepts such as classes, objects, inheritance, polymorphism, encapsulation, exception handling, collections, loops, methods, constructors, arrays, and file handling.

Although the current implementation is console-based, the project can be upgraded into a complete desktop or web application by integrating graphical interfaces, databases, and online payment systems.

PROBLEM STATEMENT

Food delivery services have become an essential part of modern life. Customers expect quick access to restaurant menus, simple ordering processes, and accurate billing.

Small restaurants and local food vendors often face difficulties in managing customer orders manually.

Traditional manual systems suffer from several limitations:

Time-consuming order management.

Increased chances of human error.

Difficulty in maintaining customer records.

Slow billing process.

Lack of organized order history.

Poor customer experience.

Therefore, there is a need for a computerized system that simplifies the entire food ordering process while improving efficiency and reducing manual effort.

EXISTING SYSTEM

In many small restaurants, customer orders are still managed manually using paper records or simple billing methods.

The existing system has several drawbacks:

Manual order processing.

Higher probability of calculation errors.

Poor record management.

No customer database.

Difficult order tracking.

Lack of automation.

Time-consuming billing.

No digital order history.

Low efficiency during peak hours.

These limitations affect both customer satisfaction and business productivity.

PROPOSED SYSTEM

The proposed Food Delivery Management System provides an automated approach for managing customer orders.

The system allows users to:

Register new accounts.

Log into the application.

Browse available restaurants.

View food menus.

Select food items.

Place orders.

Calculate total bills automatically.

Display order summaries.

Manage customer information efficiently.

The application is designed using modular programming techniques, making it easier to maintain and upgrade in the future.

FUNCTIONAL REQUIREMENTS

The application should provide the following functionalities:

User Module

User Registration

User Login

Customer Details

Restaurant Module

Restaurant List

Menu Display

Food Categories

Order Module

Food Selection

Quantity Selection

Order Confirmation

Billing Module

Automatic Bill Generation

Total Price Calculation

Order Summary

Data Management Module

Customer Information

Order Records

Basic File Handling

NON-FUNCTIONAL REQUIREMENTS

The application should satisfy the following quality requirements:

User-friendly interface.

Fast execution.

Reliable operation.

Secure data handling.

Easy maintenance.

Scalability.

Portability.

High performance.

Simple navigation.

Code reusability.

TECHNOLOGIES USED

Technology Purpose

Core Java Application Development

OOP Concepts Software Design

Collections Framework Data Storage

Exception Handling Error Management

File Handling Record Management

Eclipse / VS Code IDE

JDK Java Runtime Environment

OBJECT-ORIENTED PROGRAMMING CONCEPTS USED

The project extensively uses Object-Oriented Programming principles.

Encapsulation

Data members and methods are organized inside classes to improve security and maintainability.

Inheritance

Inheritance is used to reuse common functionalities between different classes, reducing code duplication.

Polymorphism

Method overloading and overriding provide flexibility in implementing application features.

Abstraction

Only necessary information is exposed to users while hiding implementation details.

SYSTEM ARCHITECTURE

USER

|



Login / Registration

|



Food Menu Module

|



Order Processing Module



Bill Generation



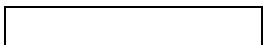
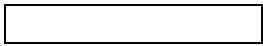
Customer Records



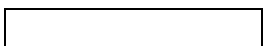
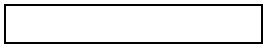
Exit



USER



Authentication Module



Restaurant Module



Order Module



Billing Module



Data Storage

LOW LEVEL DESIGN

FoodDeliverySystem

|

└─ User.java

└─ Restaurant.java

└─ Menu.java

└─ Order.java

└─ Bill.java

└─ Customer.java

└─ Login.java

└─ Main.java

MODULE DESCRIPTION

Login Module

Authenticates users before accessing application features.

Customer Module

Stores customer details and manages user information.

Restaurant Module

Displays available restaurants and food categories.

Menu Module

Shows available food items and prices.

Order Module

Processes customer selections and confirms orders.

Billing Module

Calculates the total amount and generates the final bill.

File Handling Module

Stores customer and order records for future reference.

TESTING

Testing is one of the most important phases of software development. It ensures that the application functions correctly and satisfies user requirements. The Food Delivery Management System was tested at different stages to verify its functionality, reliability, and performance.

The testing process focused on validating user input, checking the correctness of business logic, verifying menu navigation, ensuring accurate bill calculation, and handling invalid inputs gracefully.

The application was tested under different scenarios to ensure that all modules work efficiently without errors.

TEST PLAN

The primary objectives of testing were:

To verify user registration functionality.

To validate login credentials.

To ensure menu display works correctly.

To verify food ordering functionality.

To validate bill calculation.

To ensure exception handling for invalid inputs.

To check overall system stability.

TEST CASES

Test Case ID	Test Scenario	Expected Result	Status
TC-01	User Registration	User registered successfully	Pass
TC-02	User Login	Login successful	Pass
TC-03	View Restaurant Menu	Menu displayed correctly	Pass
TC-04	Select Food Item	Food selected successfully	Pass
TC-05	Place Order	Order confirmed	Pass
TC-06	Bill Generation	Correct bill generated	Pass
TC-07	Invalid Input	Error message displayed	Pass
TC-08	Exit Application	Program closes safely	Pass

TEST PROCEDURE

The application was tested using different user inputs to verify the correctness of each module.

Initially, valid customer details were entered to test registration and login functionalities. After successful authentication, different restaurants and food items were selected to verify menu operations. Bills were generated for multiple orders to ensure accurate calculations.

The application was also tested with invalid inputs to verify exception handling and input validation. All identified issues were corrected during development.

PERFORMANCE OUTCOME

The application demonstrated stable performance throughout the testing process.

The menu-driven system responded quickly to user inputs and generated bills accurately. Since the project is developed using Core Java, execution speed remained efficient without requiring significant system resources.

The modular design also makes the application easy to maintain and upgrade.

SCREENSHOTS

(Insert screenshots of your application here.)

Suggested screenshots:

Home Menu

User Registration

Login Screen

Restaurant Menu

Food Ordering

Bill Generation

Final Output

(If you do not have actual screenshots, insert placeholders with captions and replace them later.)

CHALLENGES FACED

During the internship, I encountered several technical and practical challenges while developing the project.

Some of the major challenges included:

Understanding object-oriented programming concepts.

Designing a modular application structure.

Managing multiple classes efficiently.

Implementing exception handling correctly.

Performing input validation.

Organizing project documentation.

Managing project timelines effectively.

Improving logical thinking while solving programming problems.

Each challenge helped me enhance my programming skills and gain confidence in developing Java applications.

LEARNING OUTCOMES

The internship provided valuable practical exposure to software development.

Major learning outcomes include:

Improved Core Java programming skills.

Better understanding of Object-Oriented Programming.

Practical implementation of classes and objects.

Experience with inheritance and polymorphism.

Knowledge of exception handling.

Understanding of Java Collections Framework.

Basic file handling concepts.

Software documentation skills.

Debugging and testing techniques.

Improved logical thinking.

Enhanced problem-solving ability.

Better time management.

Professional report writing.

Understanding of software development life cycle.

Overall, the internship significantly strengthened my technical foundation and prepared me for future software development opportunities.

FUTURE SCOPE

Although the current project is a console-based application, several enhancements can be implemented in future versions.

Future improvements include:

Graphical User Interface (GUI) using JavaFX or Swing.

MySQL database integration using JDBC.

Online payment gateway integration.

Food delivery tracking system.

Admin Dashboard.

Restaurant Management Module.

Customer Feedback System.

Discount and Coupon Management.

Order History.

Notification System.

Mobile Application Development.

Cloud Database Integration.

Spring Boot backend development.

REST API integration.

User Authentication using OTP.

These improvements will make the application more scalable, secure, and suitable for real-world deployment.

CONCLUSION

The Core Java Industrial Internship organized by UniConverge Technologies Pvt. Ltd. In collaboration with upskill Campus provided me with an excellent opportunity to bridge the gap between academic learning and practical software development.

During the internship, I strengthened my understanding of Core Java programming concepts and successfully applied them in developing the Food Delivery Management System using Core Java. The project enhanced my knowledge of object-oriented programming, exception handling, collections, modular programming, testing, and software documentation.

The internship also improved my analytical thinking, debugging skills, and confidence in solving programming problems. It provided valuable exposure to software development methodologies and industry practices.

Overall, this internship has been an enriching learning experience that has motivated me to continue exploring advanced Java technologies and modern software development frameworks. The knowledge and practical experience gained during this internship will contribute significantly to my future career as a Software Engineer.

REFERENCES

The following resources were referred to during the development of this project:

1. Oracle Java Documentation.
2. Java SE Programming Guide.

3. Oracle JDK Documentation.
4. Eclipse IDE Documentation.
5. Visual Studio Code Documentation.
6. GeeksforGeeks – Java Programming Articles.
7. W3Schools Java Tutorials.
8. JavaTPoint Core Java Tutorials.
9. Upskill Campus Learning Material.
10. UniConverge Technologies Internship Material.

APPENDIX

Project Modules

Login Module

User Registration Module

Restaurant Module

Menu Module

Order Processing Module

Billing Module

Customer Management Module

File Handling Module

Tools Used

Core Java

JDK 21 (or your installed version)

Visual Studio Code / Eclipse IDE

Windows Operating System

Personal Skills Developed

Problem Solving

Logical Thinking

Time Management

Self Learning

Communication Skills

Technical Documentation

Teamwork

Analytical Skills

FINAL REMARK

The successful completion of this internship has enhanced my programming knowledge, practical implementation skills, and confidence in developing software applications. I sincerely thank UniConverge Technologies Pvt. Ltd., upskill Campus, and CT Institute of Technology for providing me with this valuable learning opportunity.
